

**International Islamic University Chittagong**  
**Department Of Computer Science and Engineering**  
**Differential Equation**  
**Group- B**

**Bessel's Equation**

05

**Definition**

The definition equation  $x^2 \frac{d^2 y}{dx^2} + x \frac{dy}{dx} + (x^2 - n^2)y = 0$  is called the Bessel's equation.

**Bessel's Function**

The solution of Bessel's equation,

$$J_n(x) = \sum_{r=0}^{\infty} \frac{(-1)^r \left(\frac{x}{2}\right)^{n+2r}}{r! \Gamma(n+r+1)}$$

**Prove that,**

$$\begin{aligned} 1. J_{\frac{1}{2}}(x) &= \sqrt{\frac{2}{\pi x}} \sin x \\ 2. J_{-\frac{1}{2}}(x) &= \sqrt{\frac{2}{\pi x}} \cos x \end{aligned}$$

**Solve PDF 55**

**Legendre's Equation**

05

**1. Define Legendre's equation.** **Solve PDF 61**

**2. State Rodrigue's formula.**

- The differential equation of the form,

$$P(x) = \frac{1}{2^n n!} \cdot \frac{d^n}{dx^n} (x^2 - 1)^n$$

is called the Rodrigue's formula where n is constant.

**3. Using Rodrigue's formula find the value of  $P_0(x)$ ,  $P_1(x)$ ,  $P_2(x)$ ,  $P_3(x)$ \*\*\*,  $P_4(x)$ .**

**4. Using Rodrigue's formula, \*\*\***

Show that,  $\int_{-1}^1 [P_n(x)] dx = 0$ , when  $n \neq 0$ .

$$\int_{-1}^1 [P_n(x)] dx = 0, \text{ when } n = 0. \text{ **Solve PDF 71**}$$

### Variation of Parameter

1. Solve  $y'' + y = \operatorname{cosec} x$  **Solve PDF 80**
2. Solve  $(D^2 - 1)y = \frac{2}{1+e^x}$  **Solve PDF 81**
3. Solve  $\frac{d^2y}{dx^2} + h^2y = \sec hx$  **Solve PDF 82**
4. Solve  $(D^2 + 4)y = 4 \tan 2x$  **Solve PDF 85**

### Lagrange's Method

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1. Solve  $xzp + yzq = z^2$ \*\*\*
2. Solve  $(x^2 - yz)p + (y^2 - zx)q = (z^2 - xy)$
3. Solve  $(y + z)p + (z + x)q = x + y$
4. Solve  $(mz - ny)p + (nx - lz)q = ly - mx$
5. Solve  $(y^2 + z^2 - x^2)p - 2xyzq + 2xz = 0$

### Application of Differential Equation

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1. Bacteria in a certain culture increase at a rate proportional to the number present. If the number doubles in one hour, how long does it take for the number to triple?\*\*\*
2. When a cake is removed from an oven, its temperature is measured at 300°F. Three minutes later its temperature is 200°F. How long will it take to cool off to a room temperature of 70°F ?
3. The body of a murder victim was discovered at 11:00 p.m. The doctor took the temperature of the body at 11:30 p.m. which was 94.6°F. He again took the temperature after one hour when showed 93.4°F and noticed that the temperature of the room was 70°F. Estimate the time of death. ( Normal temperature of human body is 98.4°F ). \*\*\*
4. A generator having emf 100V is connected in series with a 10Ω resistor and an inductor of 2H. If the switch K is closed at time  $t = 0$ . Obtain a differential equation for the current and determine the current at time  $t$ . \*\*\*

**PDF 138-153**

**SURE 30MARKS**

Suggestion by

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